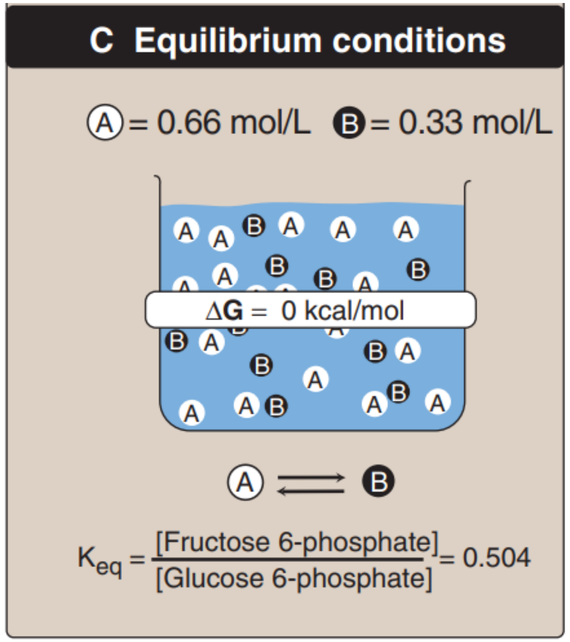
In a reaction A → B, a point of equilibrium is reached at which no further net chemical change takes place—that is, when A is being converted to B as fast as B is being converted to A. In this state, the ratio of [B] to [A] is constant, regardless of the actual concentrations of the two compounds:



$$K_{eq} = \frac{[B]_{eq}}{[A]_{eq}}$$

$$\boxed{\Delta G^{o} = -RT \ln K_{eq}}$$

$$\Delta G = 0 = \Delta G^{o} + RT \ln \frac{[B]_{eq}}{[A]_{eq}}$$

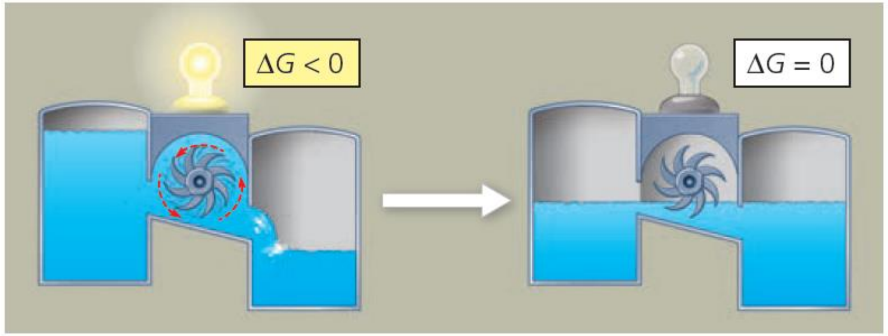
This equation allows some simple predictions:

If $K_{eq} = 1$, then $\Delta G^{o} = 0$	A $\rightleftharpoons$ B
If $K_{eq} > 1$, then $\Delta G^{o} < 0$	A $\rightleftharpoons$ B
If $K_{eq} < 1$, then $\Delta G^{o} > 0$	A $\rightleftharpoons$ B

Free energies are additive, thus a favorable reaction (ΔG1 < 0) can drive an unfavorable reaction (ΔG2 > 0), when ΔG1 + ΔG2 <0

$$
\begin{array}{llll}
(1) & \mathrm{A} \longrightarrow \mathrm{B} & & \Delta G_1'^{\circ} \\
(2) & \mathrm{B} \longrightarrow \mathrm{C} & & \Delta G_2'^{\circ} \\
\hline
\mathit{Sum:} & \mathrm{A} \longrightarrow \mathrm{C} & & \Delta G_1'^{\circ} + \Delta G_2'^{\circ}
\end{array}
$$

Figure 8.7 Equilibrium and work in an isolated hydroelectric system. Water flowing downhill turns a turbine that drives a generator providing electricity to a lightbulb, but only until the system reaches equilibrium.



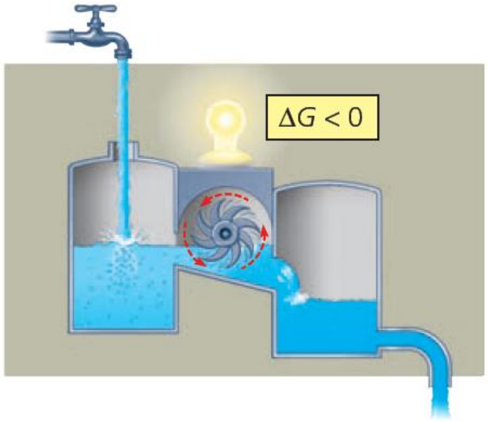
Reactions in an isolated system eventually reach equilibrium and can then do no work

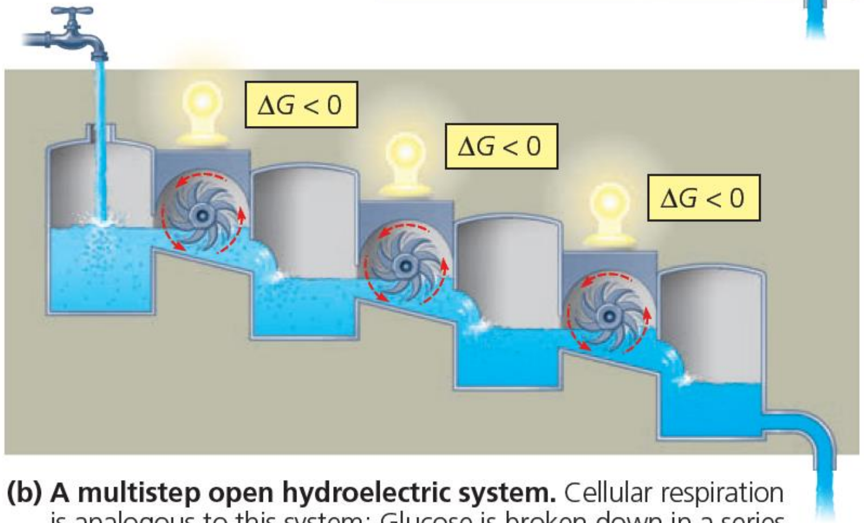
Because systems at equilibrium are at a minimum of *G* and can do no work, a cell that has reached metabolic equilibrium is dead!

The fact that metabolism as a whole is never at equilibrium is one of the defining features of life.

Figure 8.8 Equilibrium and work in open systems.

(a) An open hydroelectric system. Water flowing through a turbine keeps driving the generator because intake and outflow of water keep the system from reaching equilibrium.





(b) A multistep open hydroelectric system. Cellular respiration is analogous to this system: Glucose is broken down in a series of exergonic reactions that power the work of the cell. The product of each reaction is used as the reactant for the next, so no reaction reaches equilibrium.

ATP AS AN ENERGY CARRIER

ATP is called a high-energy phosphate compound.

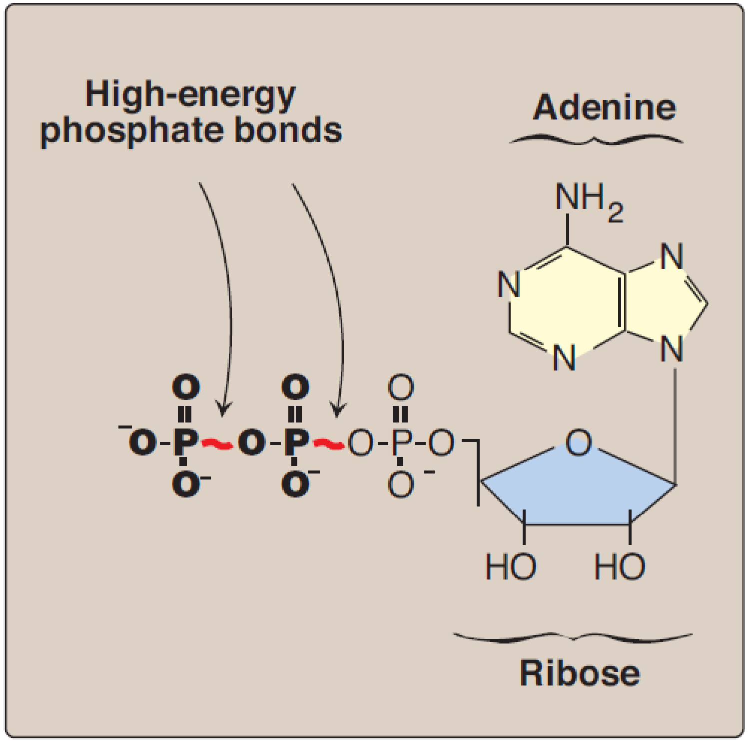
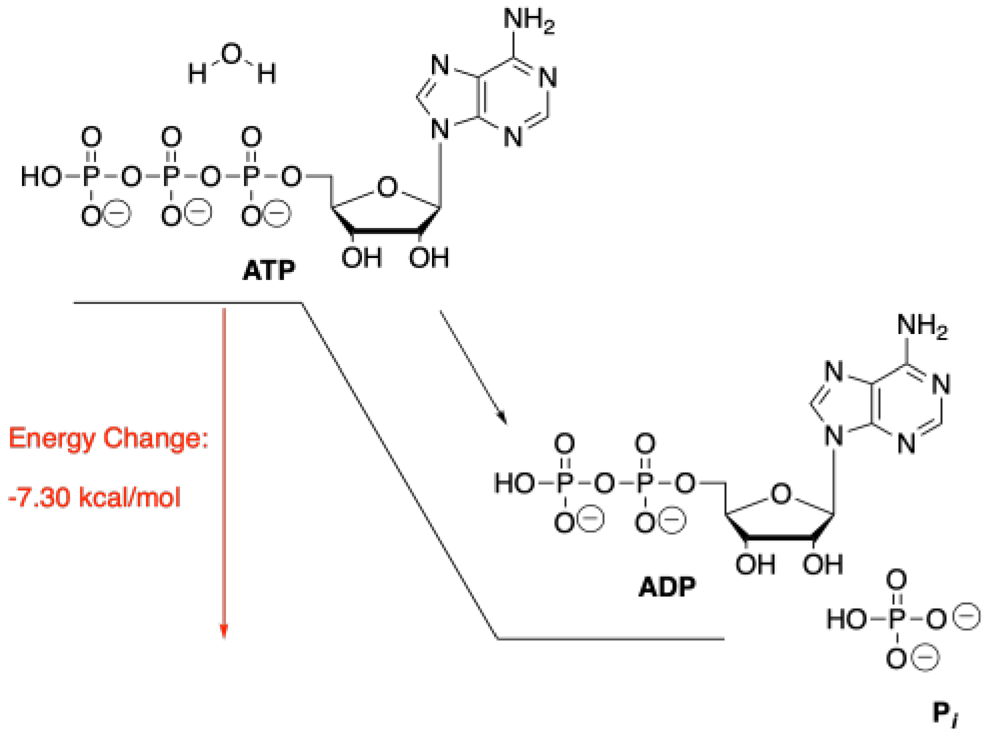


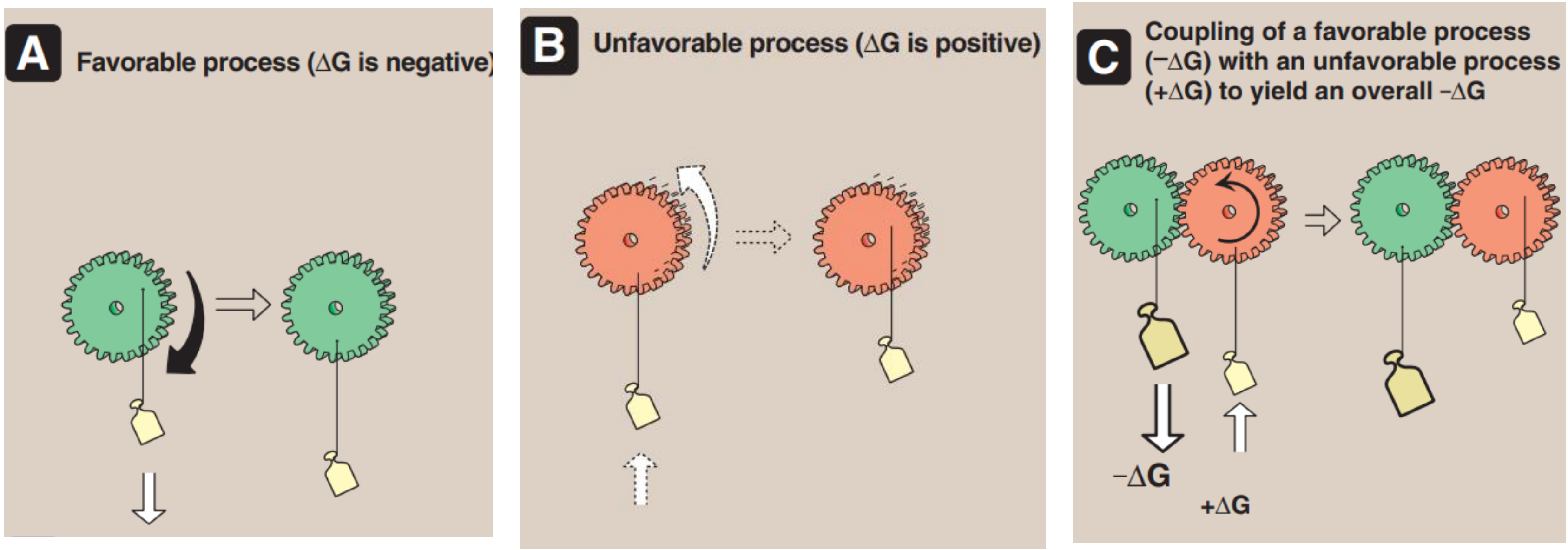
Figure 6.5
Adenosine triphosphate.



Reactions are coupled through common intermediates

$$A + B \rightarrow C + D$$

$$D + X \rightarrow Y + Z$$



many coupled reactions use ATP to generate a common intermediate.

Energetic coupling

A spontaneous (exergonic) reaction drives a non-spontaneous (endergonic) reaction.

The coupled reaction proceeds because ΔG is negative and P_i is shared between the two reactions.

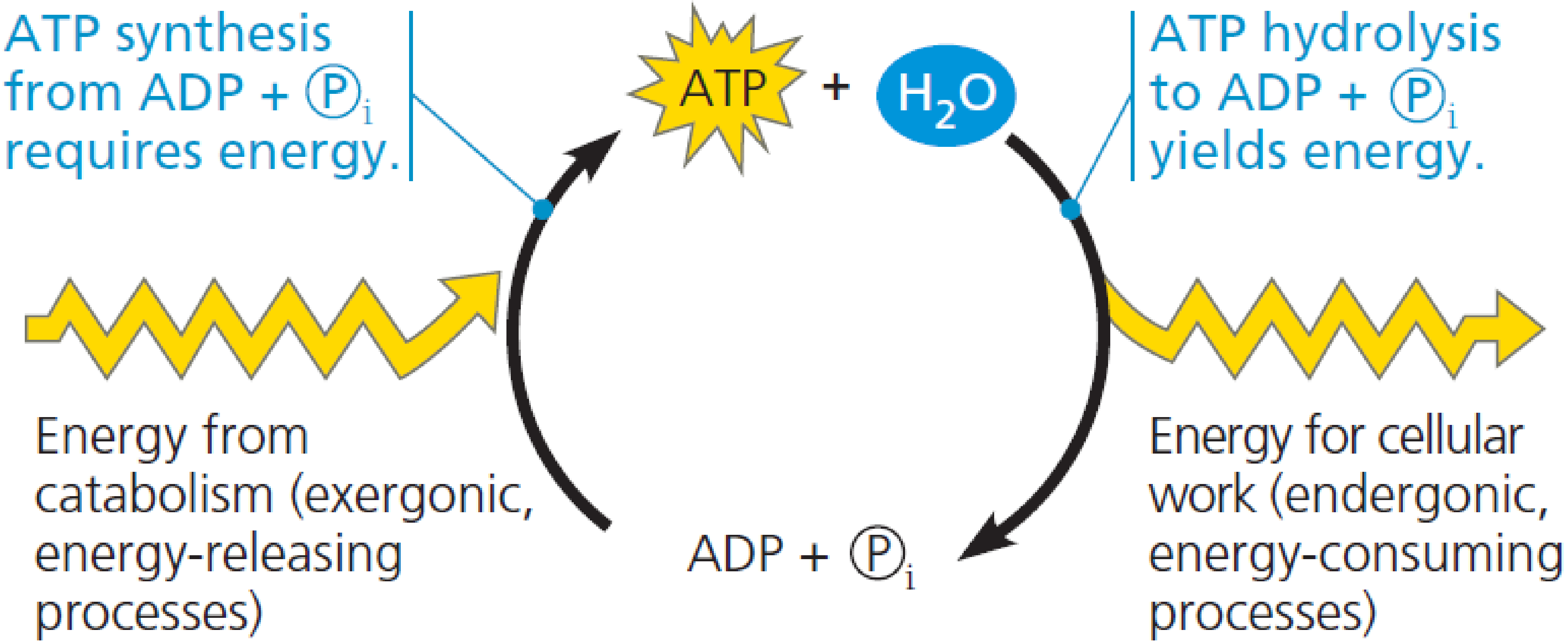
a.

ATP	+	H_2O	$\longrightarrow$	ADP	+	P_i	$\Delta G_1 = -7.3$ kcal/mol	Exergonic reaction
Glucose	+	P_i	$\longrightarrow$	Glucose 6-phosphate	+	H_2O	$\Delta G_2 = +3.3$ kcal/mol	Endergonic reaction
Glucose	+	ATP	$\longrightarrow$	Glucose 6-phosphate	+	ADP	$\Delta G = -4$ kcal/mol	Coupled reaction

b.

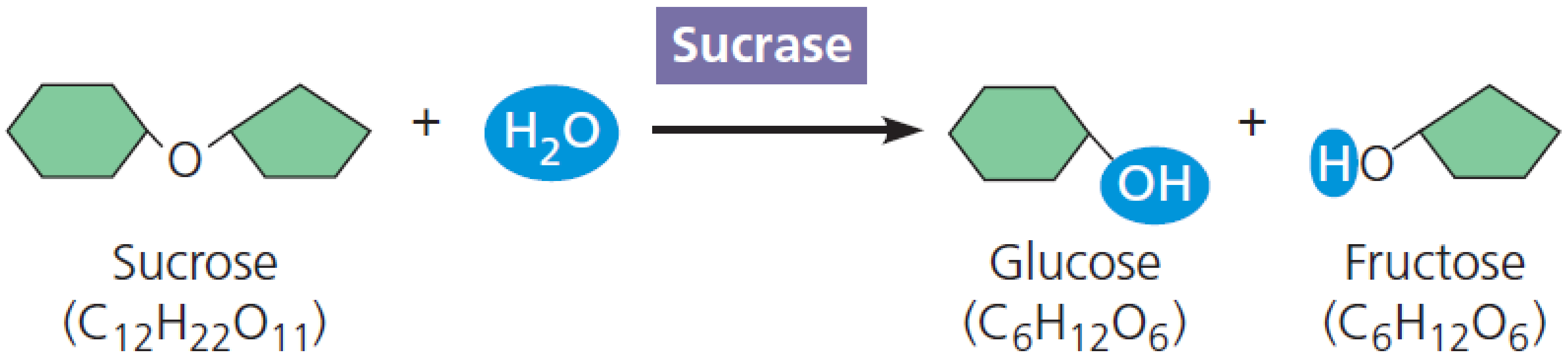
Phosphoenolpyruvate	+	H_2O	$\longrightarrow$	Pyruvate	+	P_i	$\Delta G_1 = -14.8$ kcal/mol	Exergonic reaction
ADP	+	P_i	$\longrightarrow$	ATP	+	H_2O	$\Delta G_2 = +7.3$ kcal/mol	Endergonic reaction
Phosphoenolpyruvate	+	ADP	$\longrightarrow$	Pyruvate	+	ATP	$\Delta G = -7.5$ kcal/mol	Coupled reaction

Figure 8.12 The ATP cycle. Energy released by breakdown reactions (catabolism) in the cell is used to phosphorylate ADP, regenerating ATP. Chemical potential energy stored in ATP drives most cellular work.



Enzymes speed up metabolic reactions by lowering energy barriers

A spontaneous chemical reaction occurs without any requirement for outside energy, but it may occur so slowly that it is imperceptible.
e.g. Hydrolysis of Sucrose is exergonic (ΔG=-7 kcal/mol) but takes years. Adding sucrase enzyme can complete the reaction in seconds.



How does the enzyme do this?